Globalization and Human Capital Accumulation

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Third-Year PhD Paper

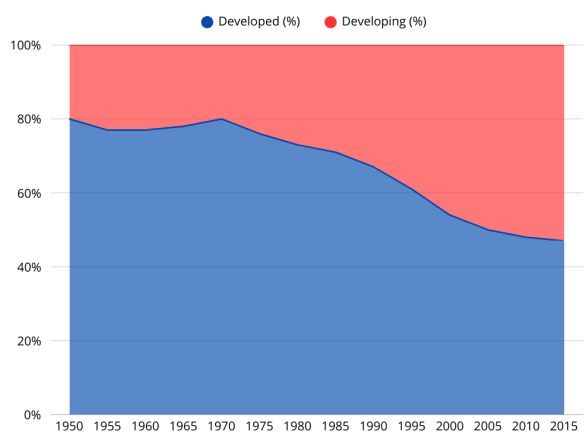
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1 Introduction

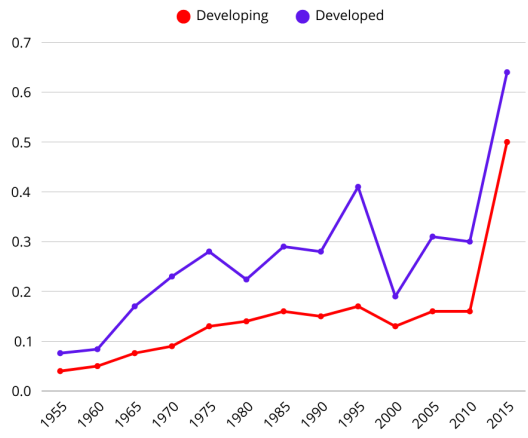
Over the past half-century, globalization has reshaped the allocation of human capital across nations. While skill-biased technical change and trade integration have raised skill premia in advanced economies (Acemoglu, 2003; Burstein and Vogel, 2017), developing countries now account for a growing share of the global educated workforce—a trend that is poorly explained by existing models. Figure 1 illustrates this paradox: the proportion of tertiary-educated individuals residing in developed countries has steadily declined since 1970, even as their absolute skill premia have remained high. However, enrollment has continued to increase in developed countries; the growing share of educated individuals in developing countries is primarily driven by their larger population base.

This paper develops a dynamic trade model with endogenous human capital accumulation to explain this divergence. I argue that specialization patterns, driven by comparative advantage and trade costs, alter the relative returns to education across countries, creating asymmetric incentives for skill accumulation that shape long-run convergence.

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(a) Tertiary: Shares in World Population



(b) Enrollment Change (pp per Year)

Figure 1: Tertiary Education and Enrollment (Based on Barro and Lee (2013))

My paper develops a dynamic Eaton-Kortum (EK) trade model with endogenous human capital accumulation, where countries differ in initial skill endowments, education opportunity costs, and sectoral productivities. Firms produce goods using skilled and unskilled labor across sectors with varying skill intensities, while households make education decisions based on skill premia. Trade patterns emerge from both Ricardian comparative advantage (differences in sectoral productivities) and Heckscher-Ohlin forces (differences in factor endowments).

The key innovation is that education enrollment responds endogenously to trade-induced changes in skill premia, creating feedback loops between specialization patterns and human capital accumulation. The model demonstrates that trade either has no effect on steady-state outcomes—while altering the speed of convergence—when countries share similar technologies, or it drives divergence when Ricardian comparative advantages dominate.

My proposed empirical strategy adapts the approach of Levchenko and Zhang (2016) to estimate sectoral productivities T_{it}^{j} from observed trade flows, wages, and skill premia. This method uses bilateral trade shares to recover technology parameters while accounting for trade costs and factor intensities. Once estimated, these parameters can be used to: (1) compute steady-state human capital stocks under zero population and technological growth

by solving the model's equilibrium conditions, and (2) simulate how trade costs affect human capital accumulation through counterfactual scenarios that vary bilateral trade barriers d^j_{nit}.

2 Literature

First, this paper relates to the literature that studies the effect of globalization and trade on the accumulation of factors of production. Different authors offer varying views on convergence between countries. The general argument is that specialization driven by comparative advantage exacerbates cross-country differences, leading to divergence in the long run—an idea consistent with this paper.

Most models predict convergence in the absence of parameter differences across countries. Oniki and Uzawa (1965) analyzes a two-sector Solow model with trade and proves conditional convergence. However, as in the standard Solow model, long-run specialization may persist if savings rates or depreciation rates differ. Subsequent work broadly follows this logic: conditional convergence, and divergence when countries differ. An important exception is Chen (1992), who shows that in a Heckscher-Ohlin model, path dependence can arise even for identical countries, leading to divergence in capital per worker. The intuition behind this result is that the Factor Price Equalization theorem ensures equal marginal returns to capital, but a country with a higher initial capital stock will continue to accumulate more capital in subsequent periods because returns remain the same. In line with this mechanism, Ventura (1997) argues that fast-growing Asian countries could accumulate more capital, as Factor Price Equalization allowed them to avoid diminishing returns to capital. However, factor price equalization does not always hold.

Building on the idea of conditional divergence, Acemoglu and Ventura (2002) shows that if countries have identical technologies, they converge to the same steady-state level of capital per worker. There are no Heckscher-Ohlin forces in the model, and within an Armington framework, countries with low capital enjoy higher returns to capital and thus

grow faster. However, capital accumulation worsens the terms of trade, reducing the rate of return until convergence is achieved. This mechanism holds in an Eaton-Kortum-type model (Eaton and Kortum, 2002) when technologies are the same. Eaton and Kortum (2001) shows that trade in capital goods facilitates technology diffusion, enabling poorer countries to import advanced machinery and accelerate productivity growth, thereby promoting conditional convergence—similar to my model. However, the steady state is determined by comparative advantage in the capital goods sector, and trade costs hinder convergence, leading to persistent specialization.

Antras and Caballero (2009) shows that financial frictions push poorer countries to specialize in labor-intensive sectors and result in capital outflows to developed countries. Similarly, Caselli and Feyrer (2007) shows that marginal products of capital are similar across countries, but capital-to-labor ratios differ due to inefficiencies and the lack of complementary endowments such as human capital.

Second, this paper contributes to the large literature on the effects of globalization on skill premia and enrollment. Atkin (2016) shows that trade liberalization increased exports in low-skill-intensive manufacturing sectors, following a significant drop in trade costs, and led to more school dropouts in the most affected regions. This mechanism aligns with my argument: higher demand for low-skill-intensive sectors lowers the return to education, resulting in reduced human capital accumulation. Using a panel of 102 countries over 45 years, Blanchard and Olney (2017) demonstrate that growth in less skill-intensive exports depresses average educational attainment, using the structure of trading partners' economies as an instrument. Other studies find increased human capital accumulation in India due to the outsourcing of high-skilled services from developed countries (Munshi and Rosenzweig (2006); Shastry (2012)).

Lastly, many papers address the rise in skill premia over the last 50 years worldwide. Several explanations are linked to globalization: the Melitz (2003) effect, where more productive firms employ more high-skilled labor (Burstein and Vogel, 2017); falling prices of

manufactured goods coupled with rising demand for skilled-labor-intensive services (Cravino and Sotelo, 2019); and globalization-induced skill-biased technical change (Acemoglu, 2003). However, these studies abstract from the accumulation of human capital. In contrast, this paper focuses on the long-run effects of globalization on skill premia through endogenous human capital dynamics.

3 Theoretical Framework

3.1 Setup

The world consists of N countries and J tradable sectors. Each sector produces a continuum of goods and is subject to trade costs. There are three factors of production: capital, high-skilled labor and low-skilled labor. Factors are mobile across sectors but immobile across countries. Trade is balanced in each period.

The flow budget constraint of the household in period t is given by:

$$P_{nt}C_{nt} = P_{nt}Y_{nt} = w_{nt}^{L}L_{nt} + w_{nt}^{H}H_{nt} + r_{nt}K_{nt}, \tag{1}$$

where P_{nt} is the price of aggregate final output, Y_{nt} is aggregate final output, H_{nt} is the high-skilled labor endowment, L_{nt} is the low-skilled labor endowment, and w_{nt}^{L} and w_{nt}^{H} are the wages of low-skilled and high-skilled labor, respectively.

The law of motion for labor (in the absence of population growth) is given by:

$$N_{nt} = L_{nt} + H_{nt} = N_n, \tag{2}$$

$$H_{n,t+1} = (1 - \varepsilon_n)H_{nt} + \varepsilon_n N_t E_{nt}, \tag{3}$$

$$L_{n,t+1} = (1 - \varepsilon_n)L_{nt} + \varepsilon_n N_t (1 - E_{nt}), \tag{4}$$

where ε_n is the birth/death rate and E_{nt} is the enrollment rate.

Aggregate final output Y_{nt} is a CES aggregate of sectoral composite goods:

$$Y_{nt} = \left(\sum_{j=1}^{J} \omega_j^{1/\eta} \left(Y_n^j \right)^{(\eta-1)/\eta} \right)^{\frac{\eta}{\eta-1}}, \tag{5}$$

where Y_{nt}^j is the composite good in sector j, η is the elasticity of substitution across sectors, and ω_j is the taste parameter for sector j.

Output in sector j, country n, and period t is produced using a CES aggregator of varieties $q \in [0, 1]$:

$$Y_{nt}^j = \left[\int_0^1 Y_{nt}^j(q)^{\frac{e-1}{e}} \, dq \right]^{\frac{e}{e-1}}, \tag{6}$$

where e is the elasticity of substitution across varieties.

The corresponding CES price index is:

$$p_{nt}^j = \left[\int_0^1 p_{nt}^j(q)^{1-e} \, dq \right]^{\frac{1}{1-e}}. \tag{7}$$

Producing one unit of good q in sector j and country n requires $\frac{1}{z_n^j(q)}$ input bundles. The cost of an input bundle is:

$$c_{nt}^j = \left((w_{nt}^H)^{\alpha_j} (w_{nt}^L)^{1-\alpha_j} \right)^{1-\beta_j} r_{nt}^{\beta_j}. \tag{8}$$

3.2 Equilibrium

Since the model does not feature forward-looking agents, I solve for a static equilibrium in each period t. A **competitive equilibrium** in this world economy consists of sequences of prices, allocations, and trade shares such that:

(i) **Firms' Optimality:** Given prices, firms choose inputs to satisfy first-order conditions and produce according to the production function.

(ii) **Households' Consumption:** Aggregate consumption is consistent with households' sectoral demand.

(iii) **Market Clearing:** Prices adjust to clear all markets—low-skilled labor, high-skilled labor, and goods.

(iv) **Balanced Trade:** Trade shares adjust to ensure trade balance in every country.

Following Eaton and Kortum (2002), equilibrium prices are given by:

$$p_{nt}^{j} = \Gamma\left(\frac{\theta + 1 - e}{\theta}\right)^{\frac{1}{1-e}} \left(\Phi_{nt}^{j}\right)^{-\frac{1}{\theta}}, \tag{9}$$

where the "multilateral resistance" term is:

$$\Phi_{nt}^{j} = \sum_{i=1}^{N} T_{it}^{j} \left(c_{it}^{j} d_{nit}^{j}\right)^{-\theta}. \tag{10}$$

Final demand across sectors is given by:

$$p_{nt}^{j} Y_{nt}^{j} = \left(w_{nt}^{L} L_{nt} + w_{nt}^{H} H_{nt}\right) \cdot \frac{\omega_j (p_{nt}^{j})^{1-\eta}}{\sum_{k=1}^{J} \omega_k (p_{nt}^{k})^{1-\eta}}. \tag{11}$$

Factor allocations satisfy:

$$\sum_{i=1}^{N} \pi_{int}^{j} p_{it}^{j} Y_{it}^{j} = \frac{w_{nt}^{H} H_{nt}^{j}}{\alpha_j} = \frac{w_{nt}^{L} L_{nt}^{j}}{1 - \alpha_j}. \tag{12}$$

Feasibility requires:

$$\sum_{j=1}^{J} L_{nt}^{j} = L_{nt}, \quad \sum_{j=1}^{J} H_{nt}^{j} = H_{nt}. \tag{13}$$

3.3 Accumulation of Human Capital

Assume each country i has infinitely many families, each deciding whether to enroll their children in college. For a family f, let V_H and V_L denote the monetary values of being high-skilled and low-skilled, respectively. Let ε_f^H and ε_f^L be i.i.d. taste shocks distributed as Type-I extreme value with mean zero and variance χ. There is also a country-specific cost of education $a_{it} = a_i + \upsilon_{it}$. To rationalize observed data, I include υ_{it}, although I set $\upsilon_{it} = 0$ for the theoretical analysis.

Each family chooses between two alternatives, and the probability of enrollment is:

$$E_f = \mathbb{P}(V_L + \varepsilon_{fit}^L \leq V_H + a_i + \upsilon_{it} + \varepsilon_{fit}^H). \tag{14}$$

Using GEV properties, the enrollment probability becomes:

$$E_{in} = \frac{1}{1 + \exp\left(\frac{V_{nt}^L - V_{nt}^H + a_n + \upsilon_{it}}{\chi}\right)}. \tag{15}$$

Intuitively, χ measures insensitivity to incentives such as the skill premium and the cost factor a_i.

Assumption 1: Assuming myopic preferences, human capital accumulation evolves according to:

$$E_{in} = \frac{1}{1 + \left(\frac{a_n w_{nt}^L}{w_{nt}^H} + \upsilon_{it}\right)^{1/\chi}}. \tag{16}$$

This follows from the utility function:

$$U_{ft} = \ln C_{ft} - \ln a_{it} \cdot \mathbb{1}(E_{ft} = 1), \tag{17}$$

where $C_f = Y_f$ is a CES aggregate of all goods. However, the subsequent results hold under any reasonable concave enrollment function.

4 Two-Country, Two-Sector Example

In the following discussion, *Ricardian comparative advantage* refers to differences in the mean productivities T_i^k across countries.

For the exercise in this section, I use parameter values from the Eaton-Kortum literature and make several simplifying assumptions:

My choice of χ is motivated by recent empirical estimates of the elasticity of enrollment with respect to returns. On average, the estimate is around 0.4 (Dynarski, 2003; Van der Klaauw, 2002; Wiswall and Zafar, 2015).

Table 1: Model Parameters and Their Values

Parameter	**Value**
ϵ	4.000
θ	8.280
η	2.000
χ	1.000
ω_j	0.500
$N_S = N_N$	1.000

4.1 Accumulation of Physical Capital

I make a Solow assumption on the accumuation of physical capital: s_n is a share of the final good that is invested in physical capital. Law of motion:

$$K_{n,t+1} = K_{nt} + I_{nt} - \delta K_{nt}$$

where I_t:

$$I_{nt} = s_n Y_{nt}$$

where Y_{nt} is final output

I assume that $s_n = 4.5\delta_n$

4.2 Identical Technologies

Consider two sectors: low-skill-intensive Agriculture (A) and skill-intensive Manufacturing (M), and two countries, North (N) and South (S). Both countries have identical average productivities ($T_N^k = T_S^k$ for $k \in \{\text{A, M}\}$), the same human capital accumulation function $E_N(x) = E_S(x)$, and equal populations ($N_S = N_N$), but differ in their initial endowment of skilled workers, with $H_N > H_S$.

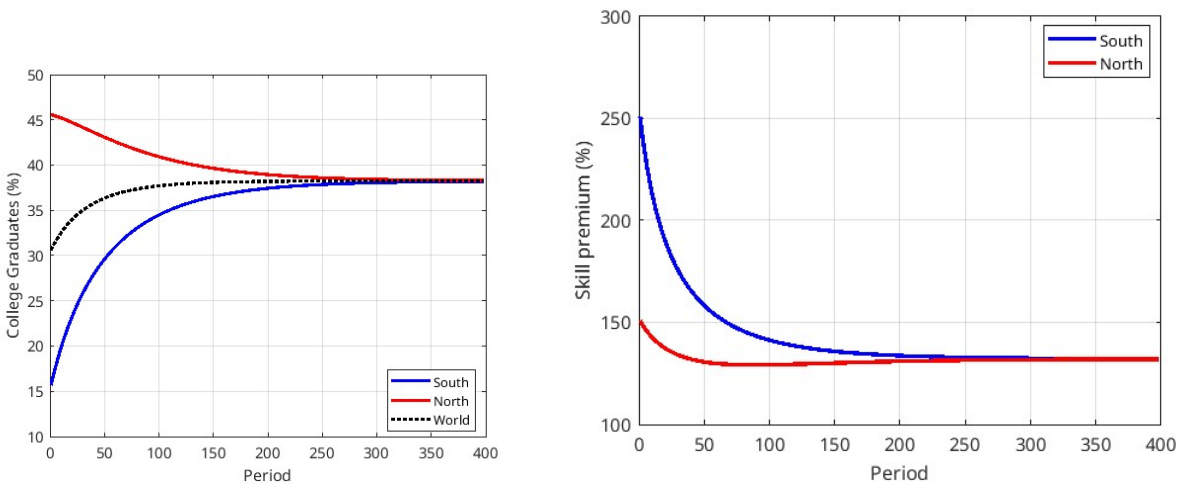
In high-income countries, the average return to four years of tertiary education is around 45%, while in developing countries it ranges from 60% to 80% (Montenegro and Patrinos, 2023). The share of tertiary-educated individuals is 47% in developed countries and 15% in developing countries (Barro-Lee data).

Technologies are symmetric and normalized such that $T_i^j = 1$ for all $i \in \{\text{N}, \text{S}\}$ and $j \in \{1, 2\}$. The cost of education is also symmetric, with $a_1 = a_2$.

Observation 1: Conditional Convergence. If countries have the same exogenous parameters, then the **steady state is not affected by trade costs**.

In autarky, the skill premium is higher in the low-skilled-labor-abundant country. After trade, the skill premium rises in the skilled-labor-abundant country and falls in the other. Unlike the standard Heckscher-Ohlin model—where factor price equalization occurs under certain conditions—the Eaton-Kortum framework allows for persistent differences: human capital remains more expensive in the low-skilled-labor-abundant country.

As a result, enrollment increases in developing countries, initiating convergence. Higher skilled labor endowment leads to lower enrollment, so countries converge until relative endowments and skill premiums equalize. In Figure 2, the North starts above steady state and the South below.



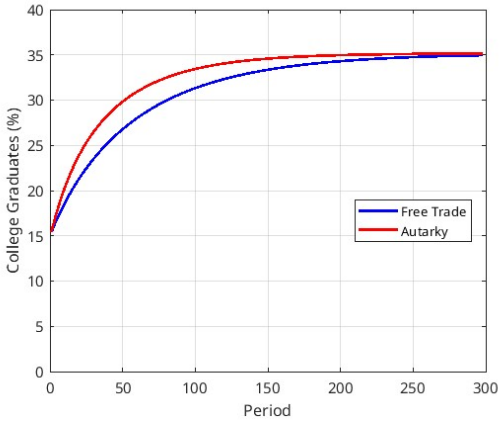
(a) Evolution of college graduates (b) Skill premium

Figure 2: Convergence dynamics

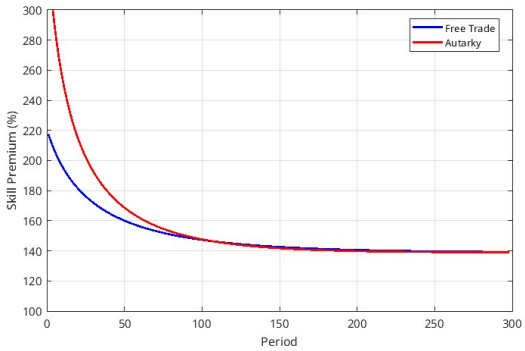
However, despite convergence in the long run, trade initially reduces the skill premium in the South, which slows convergence.

Observation 2: Slower Convergence Rate. A decrease in trade costs slows the accumulation of human capital.

This finding aligns with Galor and Mountford (2008), who argue that trade raised returns to education and enrollment in rich countries, but in poor countries, trade gains translated into higher population growth rather than more education.



(a) Evolution of college graduates



(b) Skill premium

Figure 3: Transition from autarky to trade at $t = 100$

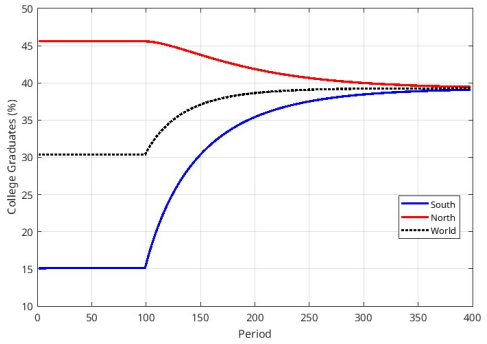
4.3 Educational Shock under Similar Technologies

Now I analyze the long-run effect of lowering educational barriers in the South. Suppose initially $a_S = 2a_N$, and then a shock occurs such that $a_S = a_N$. I focus on the long-run effects on skilled endowment and welfare in the North.

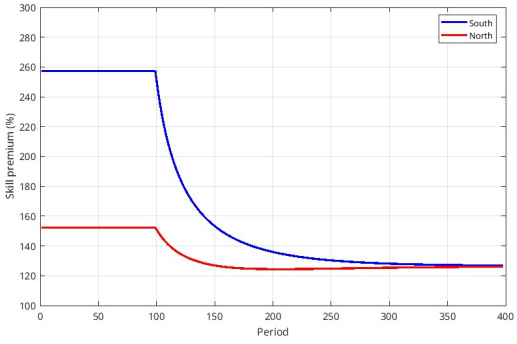
Figure 4 shows the shock at $t = 200$. Because human capital accumulation becomes cheaper in the South, its skilled labor endowment grows, which increases competition and reduces the skill premium globally.

Observation 3: Steady states depend on both domestic and foreign education costs a_i.

When education costs are higher in the South, the North has an endowment-based comparative advantage and specializes in the skill-intensive sector. Once education costs equalize, both countries converge to the same steady state—with higher college enrollment in the South and lower in the North.



(a) Evolution of college graduates



(b) Skill premium

Figure 4: Convergence following education cost shock

The effect on the Northern skill premium is non-monotonic: enrollment in the North falls, while enrollment in the South rises. Initially, the inflow of skilled Southern workers outweighs falling Northern enrollment, pushing the skill premium down. Later, the opposite occurs and the premium begins to rise again.

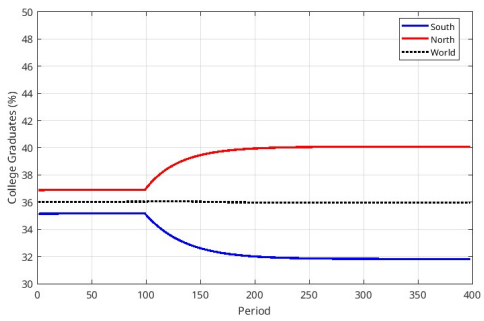
4.4 With Ricardian Comparative Advantage

Suppose all previous assumptions still hold, but now countries differ in technology: the North has a Ricardian comparative advantage in the skill-intensive sector.

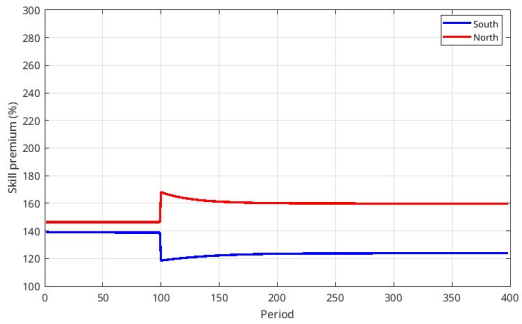
For the numerical exercise, assume $T_S^M = T_S^A = 1$, $T_N^A = 1$, and $T_N^M = 5$. In this case, specialization arises and the following result holds:

Observation 4: Divergence. The steady state is affected by trade costs. Lower trade costs widen the gap between countries.

Figure 5b shows the transition. Initially, the steady state is identical. Under autarky, the South must produce manufacturing goods, which raises the skill premium. With trade, the South specializes in agriculture, reducing the skill premium and enrollment.



(a) Evolution of college graduates



(b) Skill premium

Figure 5: Transition from autarky to trade with specialization

5 Empirical Strategy

Following the standard estimation approach in multisector Eaton-Kortum models (Chor, 2010; Costinot, Donaldson, and Komunjer, 2012; Levchenko and Zhang, 2016), I can write the following relationship:

$$\frac{\pi_{nit}^j}{\pi_{nnt}^j} = \frac{X_{nit}^j}{X_{\text{nnt}}^j} = \frac{T_{it}^j \left(c_{it}^j d_{nit}^j\right)^{-\theta}}{T_{nt}^j \left(c_{nt}^j\right)^{-\theta}}, \tag{18}$$

which, in logs, becomes:

$$\ln\left(\frac{X_{nit}^j}{X_{nnt}^j}\right) = \ln\left(T_{it}^j (c_{it}^j)^{-\theta}\right) - \ln\left(T_{nt}^j (c_{nt}^j)^{-\theta}\right) - \theta \ln d_{nit}^j. \tag{19}$$

Trade costs are modeled as:

$$\ln d_{nit}^j = d_{k,t}^j + b_{nit}^j + \text{CU}_{nit}^j + \text{RTA}_{nit}^j + \nu_{nit}^j, \tag{20}$$

where $d_{k,t}^j$ captures distance interval effects (using EK's brackets: [0–350], [350–750], [750–1500], [1500–3000], [3000–6000], [6000, ∞) miles); b_{nit}^j denotes border effects; CU_{nit}^j and RTA_{nit}^j are currency union and regional trade agreement dummies; and ν_{nit}^j is the error term. All components vary by sector j and time t.

This yields a specification similar to Levchenko and Zhang (2016):

$$\ln\left(\frac{X^j_{nit}}{X^j_{nnt}}\right) = \underbrace{\ln\left(T^j_{it}(c^j_{it})^{-\theta}\right)}_{\text{Exporter fixed effect}} - \underbrace{\ln\left(T^j_{nt}(c^j_{nt})^{-\theta}\right)}_{\text{Importer fixed effect}} - \underbrace{\theta d^j_{k,t} + \theta b^j_{nit} + \theta\,\mathrm{CU}^j_{nit} + \theta\,\mathrm{RTA}^j_{nit}}_{\text{Bilateral observables}} - \underbrace{\theta\nu^j_{nit}}_{\text{Error term}} . \tag{21}$$

To estimate the cost of education a_n, I proceed in two steps. First, I take the parameter χ from the existing literature, which estimates the elasticity of enrollment with respect to monetary incentives. Second, for the years 2005, 2010, and 2015, I calibrate a_n such that the observed change in the number of tertiary-educated individuals is matched by the right-hand side of equation (16). Finally, I take the geometric mean of the three implied values of a_n to obtain the baseline estimate used in the model.

5.1 Data

To estimate fixed effects, I can use bilateral trade flow data from the UN COMTRADE database, which provides highly disaggregated export data. However, because domestic production data is only available from 1995 onward and at a higher level of aggregation, I align my data with the 19 manufacturing sectors reported in the UNIDO Industrial Statistics Database. Detailed agricultural output is collected by the Food and Agriculture Organization of the United Nations. Total mining output can be found in the WIOD database.

To estimate relative productivity, I require data on wages and skill premia. Wage data comes from the Penn World Table 10.0 (PWT10.0), and skill premium data is taken from Montenegro and Patrinos (2023).

To examine human capital dynamics, I use data on educational attainment. I assume cross-country differences in education quality are absorbed in sectoral productivity parameters, while differences in human capital endowments are captured by the number of individuals with tertiary education. This data is sourced from the Barro-Lee Educational Attainment Dataset (Barro and Lee, 2013).

6 Quantitative Results

This section quantifies how international trade affects enrollment in tertiary education in the model. First, I present the predicted steady states for a set of countries. Then, I conduct two counterfactual exercises to evaluate these effects and to explore the interaction between physical and human capital accumulation.

6.1 Steady State

Figure 6 shows initial enrollment and the predicted steady-state levels for 15 countries. In nearly all cases, the number of educated individuals is below the steady-state level. The only exception is South Korea, which is already close to its steady state. This result is unsurprising: individuals who entered the labor force in the 1980s and 1990s—when enrollment rates were much lower—are still working today.

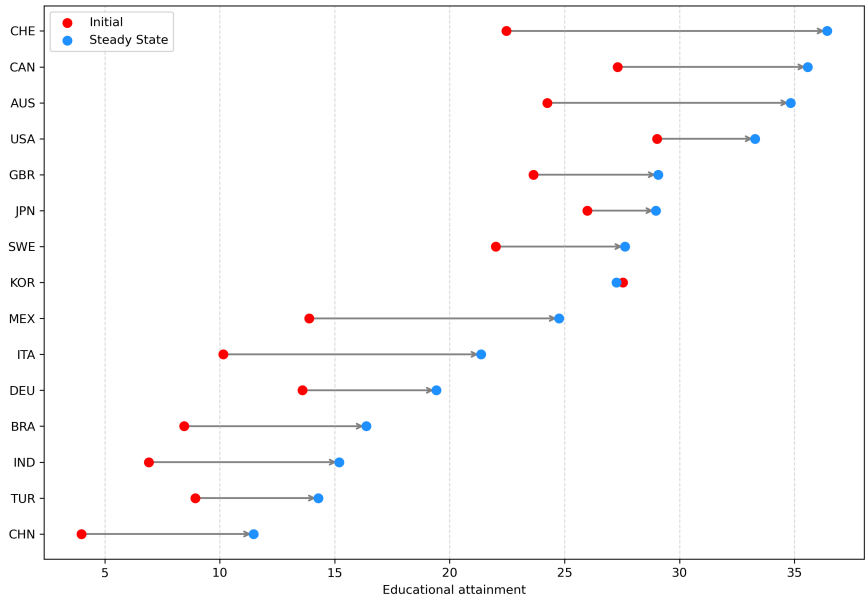


Figure 6: Initial vs. Steady-State Educational Attainment

In this model, I abstract from demographic change and migration: population is constant, and education can only be acquired domestically. Thus, in countries like China, low steady-state enrollment may reflect demographic transition effects rather than low returns

to education. In particular, the data show a slow increase in the number of educated workers in the labor force, likely because younger, more educated cohorts are smaller.

On average, the share of individuals with higher education in the labor force was 16.3% in 2015. In the steady state, this share rises to 24.3%. The total number of people with tertiary education increases by 73%, assuming no population growth.

6.2 Counterfactual 1: 30% Increase in Trade Costs

This counterfactual evaluates the impact of a 30% increase in trade costs on the steady-state level of human capital. In this model, the only channel through which trade affects skill premia is labor reallocation between skill-intensive sectors (high α_j) and low-skill-intensive sectors.

To measure the magnitude of this reallocation, I compute the following index for each country:

$$B_n = \sum_{j=1}^{J} \frac{\alpha_j \Delta L_n^j}{L_n}, \tag{22}$$

where ΔL_n^j is the change in low-skilled labor employed in sector j. Naturally, B_n should be highly correlated with changes in the skill premium and, consequently, with the steady-state level of human capital.

Figure 7 plots the change in enrollment (x-axis, in percentage points) against B_n (y-axis). As expected, reallocation is strongly correlated with changes in enrollment (correlation coefficient ≈ 0.9).

Overall, the effect on enrollment is small for most countries. The total number of educated individuals increases slightly (+0.2%) due to trade barriers. This is because education is costly, and ideally, countries with lower education costs specialize in skill-intensive sectors to conserve resources.

The largest declines in enrollment are observed in Switzerland (-2%), Ireland (-0.78%),

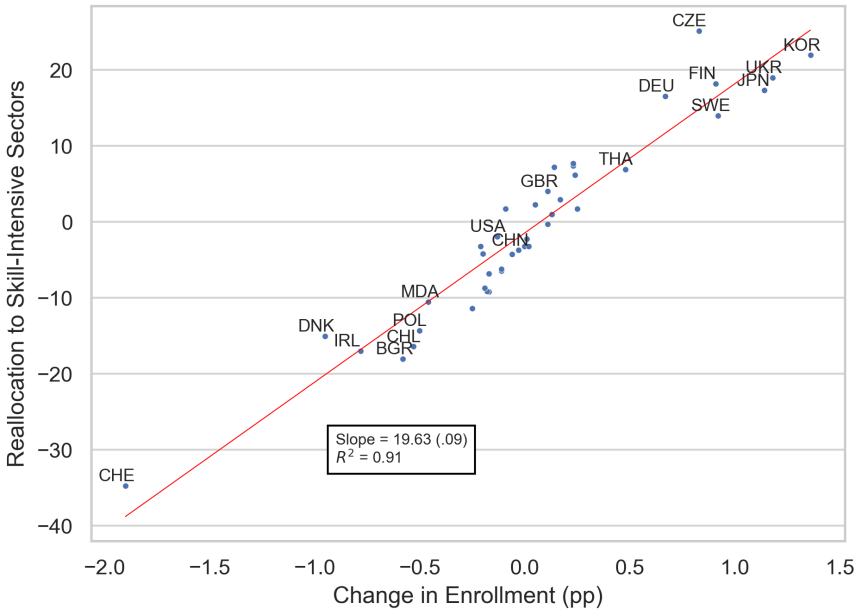


Figure 7: Change in Enrollment vs. Sectoral Reallocation Index

and Denmark (-0.95%). Countries with the greatest increases include South Korea (+1.36%), Japan (+1.14%), Ukraine (+1.18%), Sweden (+0.92%), Finland (+0.91%), and Czechia (+0.83%).

Interestingly, even developed economies experience positive effects, suggesting that some specialize in relatively less skill-intensive sectors. Higher trade barriers force them to reallocate toward skill-intensive production. This may occur because the +30% trade cost disproportionately affects highly integrated economies and disrupts specialization among close trade partners.

Denmark, Switzerland, and Ireland already have high enrollment in the baseline, while countries like South Korea and Czechia have lower initial enrollment—consistent with the convergence mechanism described earlier. Without trade, countries converge to the same steady state (if parameters are identical), but trade-induced specialization drives divergence.

While one might expect countries specialized in low-skill sectors to experience the largest increases in skill premia following higher trade costs, the effect depends on multiple conditions. First, the country must be small and highly open; otherwise, trade effects are minimal.

Second, the country must have low trade costs with a partner that has a strong comparative advantage in skill-intensive goods. In practice, most specialization (according to my model) occurs among developed countries—possibly because they face lower mutual trade costs (e.g., within the EU or Asia-Pacific trade blocs).

6.2.1 The Role of Capital Accumulation

In the model, physical capital also converges to its steady state. The capital–skill complementarity hypothesis suggests that capital accumulation may increase demand for skilled labor and thus raise enrollment. However, the Cobb-Douglas production function in this model limits the extent of this interaction. Although one might expect that capital-intensive sectors are also more skill-intensive, the empirical correlation between α_j and β_j is close to zero.

Figure 8 shows the effect of Counterfactual 1 with and without capital accumulation for a subset of countries. The overall role of capital in human capital accumulation is minimal in this specification.

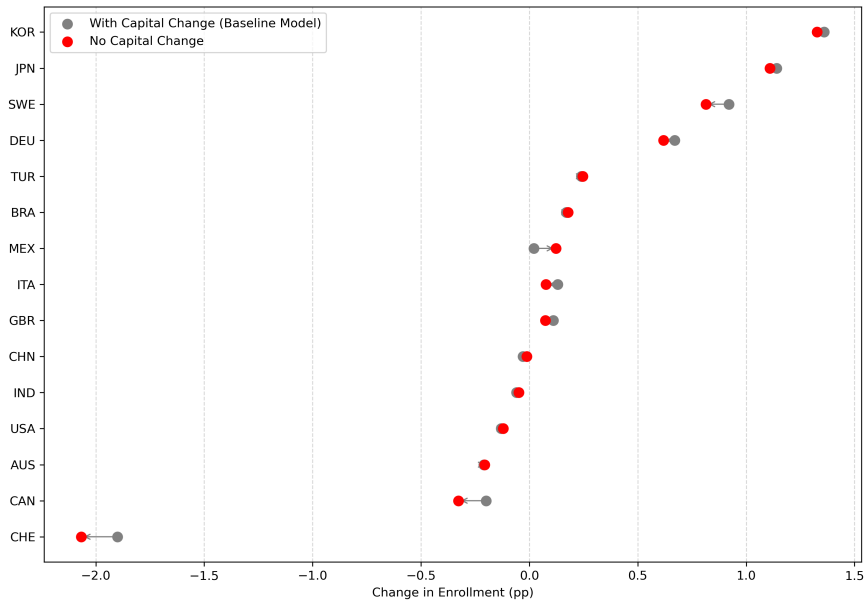


Figure 8: Counterfactual 1: With vs. Without Capital Accumulation

6.3 Counterfactual 2: Effect of China

In 2015, China had the largest population in the world (surpassed by India in 2025). As of 2024, tertiary enrollment in China reached approximately 60%, with a graduation rate of around 90%.[1] In contrast, in 2010—used in the model calibration—enrollment was about 25%. This implies that the supply of educated labor in China will rise sharply in coming years.

The model, which uses Barro-Lee data and assumes constant population, does not account for demographic dynamics. As a result, steady-state enrollment in China is just 11.4%, significantly below the global average. In this section, I explore two counterfactuals: (1) China does not accumulate human capital at all; and (2) China achieves and maintains a 60% enrollment rate, assuming the government exogenously adjusts education costs.

6.3.1 No Human Capital Accumulation in China

Suppose China's tertiary enrollment remains fixed at 3.98%. Naturally, this scenario disproportionately affects countries that are geographically close to China or have high trade shares with it.

Figure 9 shows the effect of this counterfactual on other countries as a function of their trade cost with China. The largest effects are seen in Japan, Thailand, and South Korea—China's closest and most integrated trade partners. However, even in these countries, the effect is small (less than 0.1 percentage points).

6.3.2 High Enrollment in China

Now suppose China maintains a 60% enrollment rate through exogenous policy. The global implications are similar to the previous case: despite increasing the total number of tertiary-educated individuals worldwide by 90%, the effects on other countries are small. This result reflects both the moderate influence of China on foreign skill premia in the model and the

[1] According to CEIC.

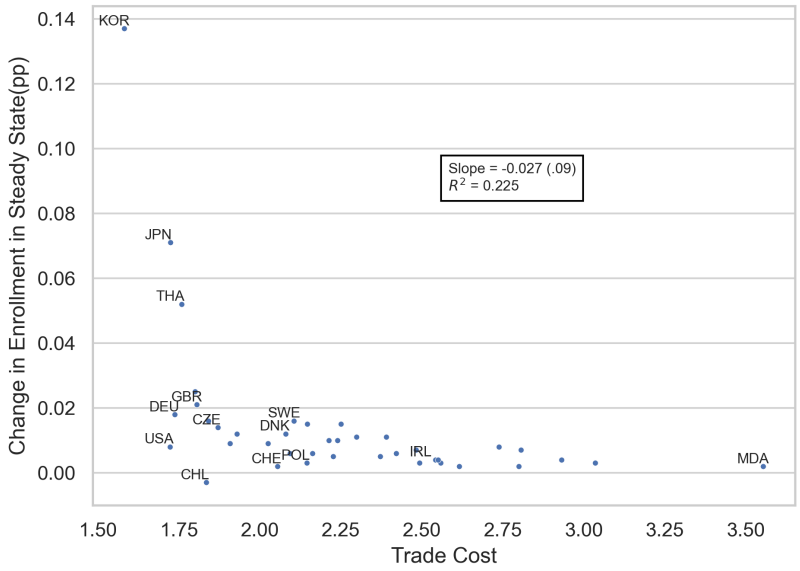


Figure 9: Steady State without Human Capital Convergence in China

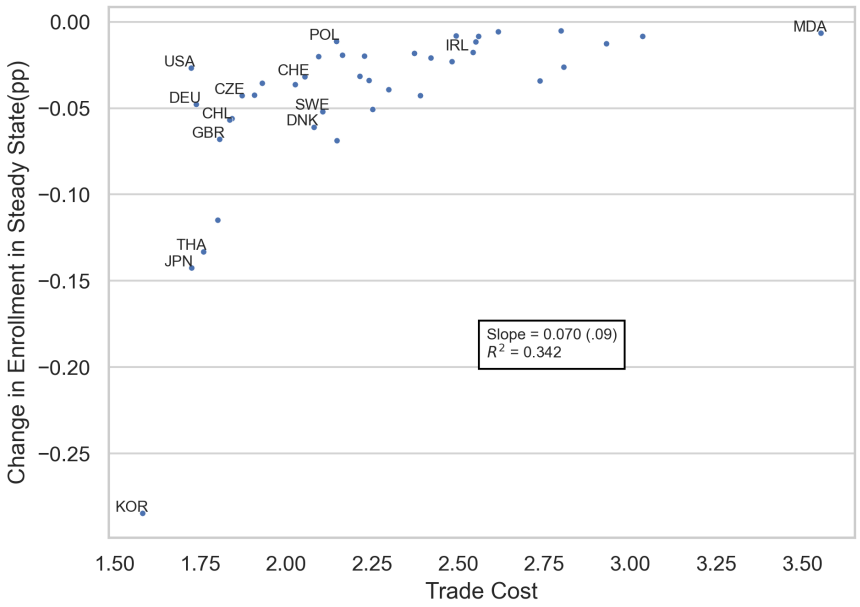


Figure 10: Steady State with 60% Enrollment in China

limited responsiveness of enrollment to changes in global returns. China incurs a welfare loss of approximately 8%, as 60% enrollment is far from its decentralized equilibrium.

Figure 10 shows the results. The impact on other countries is minor; Korea, Japan, and Thailand again show slightly larger effects (below 0.3 percentage points). In terms of welfare, all countries except Switzerland and Ireland experience small gains. The average welfare gain is 0.005%. Losses for Switzerland and Ireland are -0.01% and -0.005%, respectively.

7 Extensions

There are several avenues for extending the model:

First, the model does not include services, which may play an important role in shaping the skill premium, as discussed in Cravino and Sotelo (2019). Incorporating a non-tradable or skill-intensive service sector could amplify the effect of globalization on skill premia and increase enrollment across all countries relative to the current baseline.

Second, the model could include capital–skill complementarity directly by adopting a more flexible production function such as CES. In that case, the Solow-style assumption of exogenous savings for physical capital accumulation might be replaced with endogenous Ramsey-type accumulation for both physical and human capital.

Third, the current calibration omits agricultural and mining sectors, which may bias results for countries specializing in these sectors. Since manufacturing is relatively skill-intensive on average, including primary sectors could reveal stronger distributional effects of trade for low-skill-abundant economies.

8 Conclusion

This paper examines the long-run impact of globalization on human capital accumulation and inequality. The results show that the effects of trade depend critically on countries' initial conditions. When countries differ in technology, globalization exacerbates divergence

by encouraging low-skill specialization in developing economies. In contrast, when countries are technologically similar, trade fosters conditional convergence.

The model's key innovation is its endogenous response of educational choices to trade-induced changes in skill premia. Although globalization delivers static gains from specialization, it also modifies human capital incentives in the short run. However, these effects may not persist in the long run.

Quantitatively, a 30% increase in trade costs raises enrollment by just 0.04 percentage points on average—well below the variation driven by other factors. Similarly, China's massive human capital expansion has limited spillover effects on other countries' enrollment or welfare.

Overall, under the assumptions of the Eaton-Kortum multisector model, trade plays a modest role in shaping the global distribution of human capital. This muted effect stems from both the limited impact of trade on skill premia and the moderate elasticity of enrollment with respect to returns. As a result, globalization may be less central to human capital convergence than often assumed.

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